Rida Zahra

rz.ridazahra@gmail.com | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Computer Engineering student with practical experience and research exposure in AI/ML, Edge AI, embedded systems, Industrial IoT, DSP, and biomedical technologies. Experienced in developing intelligent systems, medical AI applications, signal-processing solutions, and hardware prototypes using Python, C/C++, ESP32, Arduino, and Raspberry Pi. Strong research interest in AI-enabled healthcare, biomedical IoT, medical imaging, assistive technologies, computer vision, and intelligent medical devices, with experience in both research and hands-on system development.

EDUCATION

NUST- National University of Science and Technology **September 2023 – June 2027**
Bachelor of Computer Engineering **(CGPA 3.94/4)** ***Summa Cum Laude*** *Islamabad, Pakistan*

RESEARCH EXPERIENCE

Research Intern **June 2026 – Present**
Sejong university x NUST *Islamabad, Pakistan*

- **AI & Foundation Models (Polyp Segmentation):** Working with BAM PolyP and related research on foundation models to improve polyp segmentation results.
- **AI & 3D Medical Imaging (Vista 3D):** Worked with Vista 3D and explored the use of foundation models for 3D medical imaging.

Independent Research **Jan 2026 – Present**

- **DSP & Audio Processing**: Compared music, religious recitation, and speech using signal processing techniques to study their effects on relaxation and calmness.
- **Haptic Feedback & Edge AI**: Researched and developed a haptic-feedback based product aimed at helping reduce anxiety and promote calmness.
- **Assistive Technology & Edge Computing**: Designed a portable communication device for people with communication difficulties, with a working prototype for voice notes and online meetings.
- **DSP & Noise Reduction**: Designed a DSP-based filter to remove unwanted noise in airport environments and developed the work into an unpublished research paper.

Final Year Research Project **August 2026 – Present**

- **Edge AI & Medical Imaging (BPPV Detection):** Developing an AI-based system to detect BPPV through nystagmus analysis, including the development of models, algorithms, and an edge-AI enabled end device.

INDUSTRY EXPERIENCE

SoC & Chip Design Intern **June 2025 – August 2025**
ChipXprt Private Limited *Islamabad, Pakistan*

- Ported and validated FreeRTOS applications on a custom RISC-V SoC; built ELF toolchain and verified ISA compatibility across RTL and QEMU emulation.
- Implemented Supervisor (S) and User (U) privilege levels, added trap routing and context switching to enable kernel/user separation — groundwork for Linux enablement.
- Performed hardware/software integration tests and regression suites (X testcases), reducing boot/trap failures during validation.

Embedded & IoT Intern **June 2026 – July 2026**
CARE Private Limited *Islamabad, Pakistan*

- Learned and worked with IoT technologies including MQTT, ESP microcontrollers, and sensor integration for industrial monitoring applications.
- Developed and tested automation programs on **Siemens S7-1200 PLCs** using Ladder Logic, implementing control sequences and interfacing with industrial hardware.
- Retrieved and monitored PLC process data by extracting runtime values for integration with external monitoring and IoT systems, supporting data acquisition and validation.

Machine Learning Intern **August 2024**
Codic Solution *Remote*

- Applied Python-based machine learning techniques to develop and optimize intelligent solutions.

SELECTED PROJECTS

ASL Alphabet Recognition | Deep CNN, TensorFlow Lite
- Trained CNN for 29 ASL classes (~99.4% accuracy); converted to TFLite for edge deployment.

Real-Time Autonomous Driving Assistant | OpenCV, YOLOv8, DIP
- Built lane/obstacle detection with real-time driving decisions; CPU-only multi-threaded pipeline (~12 FPS).

ASL Recognition on Edge Devices | ONNX, Raspberry Pi, Edge AI
- Deployed ASL model on Raspberry Pi with Firebase + Telegram Bot integration for real-time inference.

FPGA & RISC-V Chip Design | Verilog, RTL, RISC-V, FreeRTOS, QEMU, Xilinx ISE
- Designed 5-stage pipelined RISC-V processor (forwarding, hazard detection), 16-bit processor, and 4-bit ALU. Ported FreeRTOS to a custom RISC-V SoC and extended privilege levels (M/S/U) for Linux enablement; validated on QEMU/FPGA.

CalmSync – ESP32 Stress Wearable | ESP32, C++, BLE, IoT
- Prototyped wearable stress-monitoring device with haptic feedback; validated on Wokwi and hardware.

Airport Noise Speech Enhancement | MATLAB, FIR/Wiener Filtering, DSP
- Built two-stage speech enhancement system; achieved 2.28 dB mean SegSNR improvement on NOIZEUS corpus.

5-Band Graphic Equalizer | MATLAB, Simulink, DSP
- Built real-time equalizer with Butterworth filters, per-band gain control, and interactive GUI.

Single-Stage Audio Amplifier | BJT, Analog Circuit Design
- Designed common-emitter amplifier achieving ≥50x voltage gain, driving 8Ω/16Ω speaker load.

TECHNICAL SKILLS

- **Programming:** Python, C++, C, Java, Verilog (FPGA/RTL), RISC-V Assembly
- **Artificial Intelligence & Machine Learning:** TensorFlow, TensorFlow Lite, Scikit-learn, ONNX, Deep Learning Fundamentals, Edge AI Deployment
- **Computer Vision & Autonomous Systems:** OpenCV, YOLOv8, Real-Time Object Detection, Lane Detection, Multi-Stream Video Processing
- **Software & System Development:** Git, REST APIs, ASP.NET Core MVC, Bootstrap, Linux Development
- **Embedded AI & IoT:** ESP32, Raspberry Pi, Arduino, Embedded Linux, FreeRTOS
- **Engineering Tools:** MATLAB, Simulink, Proteus, Wokwi, QEMU

AWARDS & LEADERSHIP

- Vice President BRISC (Research and Innovation Society)
- Winner NSTP Pitch Bazar
- Winner Hult Prize (Campus Round)
- 1st Runner-Up COMPPEC 2025 - nationwide technical idea pitching competition
- 2nd Runner-Up Zindagi Prize – ideathon for business ideas

REFERENCES

Dr. Muhammad Usman Akram
Professor & Head, Department of Computer & Software Engineering
NUST College of Electrical & Mechanical Engineering (CEME), Rawalpindi, Pakistan
Email: usmakram@gmail.com

Dr. Muhammad Zubair Islam
Assistant Professor, Department of AI and Robotics
Sejong University, Seoul, South Korea
Email: mzislam@sejong.ac.kr